# INTRODUCTION

As introduced in Alternatives Learning Module 1, digital assets are a relatively new investment class that covers assets that can be created, stored, and transmitted electronically and have associated ownership or use rights. This class includes a wide variety of digital assets, including cryptocurrencies, tokens, and digital collectables. Based on the innovative distributed ledger technology (DLT), or blockchain technology, digital assets utilize advanced encryption techniques that assure the authenticity of digital assets. While cryptocurrencies have their own blockchains, crypto-tokens are built on an existing blockchain.

As with other types of alternative investments, digital assets have characteristics distinct from traditional investments. Since the advent of Bitcoin in 2009, when it was a niche concept in the technology world, digital assets have gradually become more mainstream alternative investments for investors worldwide. We will give an overview of the common forms of digital asset investments and discuss the key concepts and pitfalls in this space.

Despite their special features and technological characteristics, digital assets offer investors diversification while providing higher expected returns than traditional investments provide. However, their risks are also higher. We will discuss the sources of risks in digital assets to provide a better understanding of this evolving asset class.

## Learning Module Overview

- Blockchain and distributed ledger technology might offer a new way to store, record, and track digital assets on a secure, distributed basis. Additionally, DLT could bring efficiencies to post-trade and compliance processes through automation, smart contracts, and identity verification.
- DLT can take the form of either permissionless or permissioned networks.
- A consensus protocol is a set of rules that govern how blocks are cryptographically chained together in a blockchain network for the verification of the complete and immutable history of transaction records. Two broad types of consensus protocols are “proof of work” (PoW) and “proof of stake” (PoS).
- Digital assets are frequently seen as an alternative asset class. As digital assets become more developed, institutional investors may continue to seek some exposure to these assets for their higher returns and possible diversification benefits.
- Digital assets differ from traditional financial assets in terms of their inherent value, transaction validation approach, uses as a legal medium of exchange, and legal and regulatory protection.
- The most common digital assets are cryptocurrencies, including Bitcoin and altcoins (including stablecoins and meme coins). There are also digitalized tokens that include non-fungible tokens, security tokens, utility tokens, and governance tokens.
- Many cryptocurrencies are designed with self-imposed limits on the total supply through complex computer algorithms. Such limits could help maintain a value from a technical perspective, yet there is no economic consensus on how they should be valued. Cryptocurrency exchanges are classified into centralized exchanges and decentralized exchanges. Both centralized and decentralized exchanges face

- Investment in digital assets can take the form of direct ownership of cryptocurrencies and other digital assets on the blockchains or indirect investment in exchange-traded products, hedge funds, trusts, futures, and thematic stocks.
- Asset-backed tokens are digital claims on physical assets, financial assets, or financial instruments and are collateralized by these underlying assets.
- The push for financial decentralized applications based on open-source codes and smart contracts has grown into a movement known as decentralized finance, or DeFi. DeFi seeks to design, combine, and develop decentralized financial applications as building blocks for sophisticated financial products and services.
- The price of Bitcoin and other digital assets are driven by expectations on future asset appreciation rather than any underlying cash flow. The market demand for the limited supply of cryptocurrencies is a significant driver of prices.
- The performance of the first widely traded digital asset, Bitcoin, has been characterized by high return, high volatility, and low correlations with traditional asset classes.
- Due to the historically low correlations with other asset classes, digital assets offer potential diversification benefits to a well-diversified portfolio. But the correlations are observed to have risen, especially during periods of high market uncertainty.

## Learning Module Self-Assessment

These initial questions are intended to help you gauge your current level of understanding of this learning module.

**1.** The consensus protocol on a distributed ledger technology network refers to:

**A.** the standardized approach that governs how digital assets generated from a blockchain network should be valued.
**B.** programs embedded in electronic transaction records that are coded to self-execute according to predetermined conditions.
**C.** the set of rules governing how blocks are cryptographically linked to the chain to become immutable on the distributed ledger network.

Hide Solution

**Solution:**

The correct answer is C. When transactions enter a node of the distributed ledger, they are bundled into “blocks” and cryptographically “chained” together to facilitate verification of the prior history. How blocks are chained together is determined by the consensus protocol, a set of rules governing how blocks can join the chain and become the immutable “truth.” The consensus protocols are designed to resist attempts at malicious manipulation up to a certain level of security.

**2.** Cryptocurrencies are a common form of digital asset that:

**A.** can be mined only through “proof of work” on blockchain networks.
**B.** exist both in physical forms and electronic records with rights to use, buy, or sell by owners.
**C.** can be used to transfer or store value, which allows time-efficient transactions between parties without the need for an intermediary.

Hide Solution

**Solution:**

The correct answer is C. Cryptocurrencies are used to transfer or store value, which allows near-real-time transactions between parties without the need for

Depending on the particular consensus protocols used, cryptocurrencies can be generated either by “proof of work” or “proof of stake” by miners or validators on networks.

**3.** Digital assets differ from traditional financial securities in that:

**A.** digital assets are subject to a broader legal protection framework.
**B.** digital assets are not valued based on their expected future cash flow.
**C.** transaction records of digital assets require a centralized intermediary.

Hide Solution

**Solution:**

The correct answer is B. Most digital assets do not have a fundamental value based on underlying assets or on the potential cash flow or earnings they are expected to generate. To date, the legal and regulatory frameworks for digital assets are still evolving, and there is generally less legal protection compared to traditional financial securities. Digital asset transactions are recorded on a distributed ledger, and no centralized intermediary is required in the process.

**4.** A special type of cryptocurrency that is backed by and pegged to a fiat currency is called:

**A.** altcoin.
**B.** stablecoin.
**C.** meme coin.

Hide Solution

**Solution:**

The correct answer is B. A stablecoin is designed to maintain a stable value by pegging its value to another asset and is collateralized by a basket of assets, typically a fiat currency, precious metals, or other cryptocurrencies.

**5.** An indirect investment in digital assets can be made through:

**A.** entering into cryptocurrency futures.
**B.** purchasing Bitcoins on cryptocurrency wallets.
**C.** participating in the initial coin offering of a new digital token.

Hide Solution

**Solution:**

The correct answer is A. Investors interested in an indirect investment in digital assets can trade cryptocurrency futures on established exchanges, such as the Chicago Mercantile Exchange. However, the purchase of Bitcoins and buying new tokens through an initial coin offering are direct forms of investment.

**6.** A cryptocurrency ETF seeks to:

**A.** gain exposure to cryptocurrencies through cash and cryptocurrency derivatives.
**B.** replicate digital asset investment returns by investing directly in cryptocurrencies.
**C.** gain exposure to the cryptocurrency theme by investing in public equities related to the digital asset sector.

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**Solution:**

The correct answer is A. A cryptocurrency ETF seeks to replicate digital asset investment returns by cash and cryptocurrency derivatives.

**7.** Bitcoin as an alternative investment has historically exhibited:

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C. high return, high risk, and high correlations with traditional assets.

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**Solution:**

The correct answer is B. Despite no future cash flow generation, Bitcoin as an alternative investment has been characterized by high return, high volatility, and low correlations with traditional asset classes.

**8.** Early investors in cryptocurrencies have enjoyed significant price appreciation but many later-stage investors suffered huge losses most likely because:

**A.** blockchain technology favors early-stage investors.
**B.** there are substantial price fluctuations in cryptocurrencies.
**C.** cryptocurrencies have increased their appeal to institutional investors.

Hide Solution

**Solution:**

The correct answer is B. Because cryptocurrencies are a relatively new innovation, their market is subject to rapid price swings, changes, and uncertainty. Their prices have been extremely volatile over the years, and price drawdowns were often much more substantial than for traditional asset classes. As a result, later-stage investors may suffer huge losses depending on their points of entry and exit.

Discuss Filter

Discussion

What do you want to discuss... (required)

SP

Q7: Bitcoin as an alternative investment has historically exhibited:
low return, low risk, and high correlations with traditional assets. X
high return, high risk, and low correlations with traditional assets. In theory
high return, high risk, and high correlations with traditional assets. Market Norm

Though in theory BTC is low correlation to trad. Nowadays it's more correlated that ever tbh.

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